

Exploring a Design Science approach to visualizing complex automotive software architectures

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Abstract

Modern automotive software architectures have evolved into highly complex networks of software components, electronic control units, signals, and thousands of properties. Understanding these multi-format, large-scale, highly-nested software structures is a significant cognitive challenge for domain experts. To address this, this Work-in-Progress paper presents an interactive software visualization artifact developed following the Design Science Research Methodology. The designed artifact ingests heterogeneous automotive vehicle data (ARXML, YAML, and binary files) and establishes a unified pipeline to transform these raw, disconnected metadata into comprehensive network models. By applying knowledge graph principles, the system explores graph visualization libraries to build stable mental models of complex software structures. Finally, this study proposes to evaluate the artifact through a comprehensive usability study with domain experts, while considering its support for users' cognitive mapping.

CCS Concepts

• **Human-centered computing** → **Visualization toolkits; Knowledge Graphs; Empirical studies in visualization.**

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1 Introduction and Theoretical Background

The automotive industry is undergoing a major transformation toward software-defined vehicles, where software is decoupled from hardware to enable continuous updates, adaptability, and improved performance [15]. This shift, aligned with Industry 4.0, has turned vehicles into data-driven “digital platforms on wheels,”

with value increasingly derived from connectivity and continuous data exchange rather than purely mechanical design [2].

However, this transformation introduces significant challenges, particularly a “data tsunami” caused by rapidly growing sensor-generated data, expanding software complexity, and extensive data exchange across supply chains [3, 19]. In addition, another key challenge is the increasing functional data complexity in industry-standard software systems [22]. A popular automotive software format, ARXML (AUTOSAR Extensible Markup Language¹) files produce accurate, standardized representations of a system, but it is not readily comprehensible to humans [22], because they have a deeply nested structure with many levels in its hierarchy.

As a result, engineers are heavily reliant on complex configuration tools like DaVinci Configurator² or specially designed parsing systems to model component interactions [22]. However, they still fail to present data in a semantically meaningful, human-readable way. Instead, the data from the configuration tools is displayed in separate, disjointed tables or high-density nested trees rather than establishing semantic relationships [20]. This creates a research gap in transforming machine-readable data into cognitively accessible representations.

1.1 Design Science in Human-Centered Computing

Design science leverages prototyping and iterative development to build tangible solutions categorized into **constructs, models, methods, and instantiations** [14]. To use this constructive approach, Peffers et al. [24] established the Design Science Research Methodology (DSRM), a structured, iterative six-stage workflow encompassing **problem identification, objective definition, design and development, demonstration, evaluation, and communication**. This study applies DSRM to iteratively design, implement, and evaluate an interactive knowledge-graph visualization artifact, transforming complex, multi-format vehicle metadata into an intuitive, system-wide mental model for domain experts. Consequently, rather than serving as a linear sequence, DSRM functions as a flexible methodology to bridge raw machine-readable data and human-centered interfaces.

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¹<https://autosar.io/en/insights/arxml-autosar-xml-guide>

²<https://www.vector.com/int/en/products/products-a-z/software/davinci-configurator-classic/>

1.2 Knowledge Graphs and Cognitive Mapping

Knowledge graphs serve as an effective socio-technical approach for modeling, integrating, and interpreting complex, highly interconnected information across heterogeneous datasets [18, 26]. This capability is uniquely applicable to the software-defined vehicle domain, where massive volumes of complex configuration data are continuously generated and deeply nested across multiple subsystems [6]. By mapping these abstract, multi-format datasets into a relational topography, knowledge graphs translate machine-readable infrastructures into human-interpretable models.

Ultimately, because this graph-based architecture fundamentally mirrors the relational way humans naturally organize knowledge, we believe it aligns directly with cognitive science principles such as cognitive mapping [28], thereby it may enhance user comprehension and usability during system-wide troubleshooting. A fundamental premise of this study is the structural and functional alignment between human cognitive maps and knowledge graphs, as both represent discrete entities and their interrelationships to support navigation, inference, and the organization of interconnected information [8, 16]. As the relational topology of knowledge graphs directly mirrors these human cognitive structures, graph-based visual representations serve as an ideal computational solution that may enhance knowledge discovery and support human reasoning when navigating complex data environments [8].

1.3 Related Work

While prior studies successfully use knowledge graphs to manage complex data [6, 8, 10], a notable gap remains in applying these visual representations to the deeply nested, large-scale data structures native to software-defined vehicles [3, 17]. Notably, Fahd et al. [8] used DSRM to develop a knowledge graph that transforms complex dementia research data into clear node-and-edge relationships. Their study applies structured cognitive scripting to help medical researchers identify hidden connections in scientific texts [8]. Consequently, the study shows that explicitly visualizing relationships serves as an external aid to human understanding, enabling experts to navigate dense information more easily and make faster structural inferences.

Meanwhile, most existing work focuses on technical system analysis [10], with less emphasis on how such representations align with human cognitive processes or how they support interpretability beyond expert users. In addition, while some approaches address large-scale or dynamic data environments [6, 17], they rarely consider highly interconnected and deeply nested structures, as is the case with software-defined vehicle data.

1.4 Contribution and Research Question

To bridge this research gap, this study proposes a knowledge-graph-based approach that proposes the visualization of vehicle data as semantic networks for better usability aiming towards the formation of cognitive maps in complex systems using a design science research methodology (DSRM), in collaboration with Volvo Cars³. This research makes the following contributions: **1)** A unified Python pipeline that parses, normalizes, and links heterogeneous

automotive formats (ARXML, YAML, and binary data) into a central relational network; **2)** The design and implementation of a web-based, highly interactive knowledge graph visualization interface built using *React* and *ReactFlow*; and **3)** The proposal of an empirical evaluation with domain experts performing real automotive engineering tasks, comparing the developed artifact against a standard legacy baseline.

We conduct this research while posing and exploring the **research question**: *How can DSRM be used to enable transforming complex, highly-nested machine-readable data from the automotive industry into interactive knowledge graphs?*

2 Research Methodology

2.1 System Requirements from Design Science

The primary objective of this study is the creation and evaluation of innovative artifacts designed to solve identified organizational problems. To develop the design solution, this study uses the four distinct, interconnected design classifications established in Design Science, by Hevner et al. [14]:

1) Constructs: The specialized vocabulary and conceptual definitions (such as ECUs, PDUs, Signals and properties) established to structure and talk about complex automotive datasets.

2) Models: The semantic knowledge graph schema or concept model, that formally defines how these concepts are interconnected to represent the overall system topology.

3) Methods: The underlying parsing algorithms and data-transformation pipelines written to automatically extract raw configuration files and map them into the graph-based representation.

4) Instantiations: The functional web visualization interface which implements the underlying models and methods into a working system, enabling engineers to trace domain constructs directly from raw metadata.

Together, these four interdependent types ensure that the final research output delivers a practical solution for industry challenges and theoretical insights for academia [14].

2.2 Design Science Research Methodology

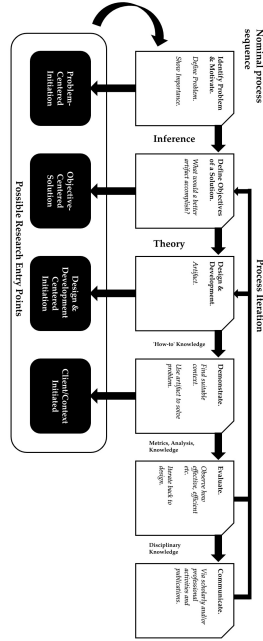
For the research output, which is a design artifact, this study adopts the Design Science Research Methodology (DSRM) framework proposed by Peffers et al. [24]. DSRM provides a structured approach for producing innovative artifacts to solve defined problems with multiple iteration [24]. Table 1 illustrates the application of the six-stage DSRM cycle.

These six stages are not a strictly rigid, linear sequence. Rather, they form an inherently iterative lifecycle of construction and evaluation [12, 24], as illustrated in Table 1. Through these continuous feedback loops—such as returning from evaluation to design revisions—researchers can effectively build, test, and refine innovative technological solutions to address complex, real-world engineering challenges [27]. For example, observations gathered during the evaluation phase (stage 5) are fed back into the design and development phase (stage 3) so that the interactive system can be iteratively modified based on data.

³For corporate confidentiality requirements, all specific signal names, software components, and architecture attributes visible within this paper have been replaced with

mock data. This transformation preserves the structural topography and technical complexity of the network while ensuring proprietary data security.

Table 1: The DSRM Process Model, highlighting the iterative feedback loops between evaluation and design revisions. Image Source: Barry et al. [11].



DSRM Stage	Objective & Operational Context
1) Identify problem and motivate	Highlight cognitive overload when domain experts manually navigate fragmented, multi-format automotive data (ARXML/YAML).
2) Define objectives of a solution	Define technical requirements: develop a unified ingestion pipeline and interactive relational visualization.
3) Design and development	Construct the artifact: build a Python extraction engine backend with a <i>React</i> , <i>ReactFlow</i> , and <i>Cytoscape.js</i> frontend.
4) Demonstrate	Deploy the functional visualization engine using complex, randomized configuration data to prove feasibility.
5) Evaluate	Conduct empirical user testing across specialized tasks, tracking SUS metrics.
6) Communicate	Diffuse research contributions via academic publishing and internal reports.

3 Research Process

3.1 DSRM Stage 1 - Identifying Issues in the Current Interface

To identify current challenges, it is first necessary to evaluate the limitations of the company-specific configuration tool used by engineers to navigate datasets of software-defined vehicles. Based on exploratory observations and an analysis of structural data bottlenecks, three primary systemic difficulties were identified within existing interfaces:

1) **Data Fragmentation across disjointed views:** The current system isolates complex datasets, requiring engineers to navigate critical properties through rigid tabular views or deeply nested hierarchical structures. As a result, engineers must manually cross-reference multiple windows to trace a single system behavior.

2) **Inability to expose source code metadata:** The proprietary, in-house data configuration interface focuses more on showing strict data parameters and values than on helping users explore and understand the full range of data. As a result, valuable contextual metadata embedded within the underlying vehicle source code (e.g., ARXML) remains hidden or difficult to access within the interface.

3) **Obscured end-to-end signal and property flow:** One of the most serious operational bottlenecks is how difficult it is for engineers to trace end-to-end communication flows. Since the current interface relies on static and disconnected text lists instead of explicit semantic links, tracing this end-to-end lineage through the system is extremely laborious and error-prone.

3.2 DSRM Stage 2 - Objectives

To address these challenges, this study proposes a knowledge-graph-based visualization artifact developed in accordance with the six stages of DSRM (Table 1). The resulting artifact ensures alignment with foundational Design Science artifact types (Section 2.1), serving not only as a practical tool that enhances usability and supports engineers' understanding and efficiency, but also as a theoretical contribution to the domain of automotive data visualization research.

3.3 DSRM Stage 3 - Artifact Design and Development

Knowledge-graph-based Design: The initial architectural design was motivated by **Sankey diagrams** and **parallel coordinates**. Sankey diagrams effectively illustrate directional flows and structural dependencies [21], while parallel coordinates are well suited to represent multidimensional relationships across distinct axes [5]. However, we found that these graph types are limited in representing the deeply nested, non-linear dependencies characteristic of automotive software systems, although they present sequential or linear transitions effectively.

Thus, this study adopts knowledge graphs as the primary visualization method, as they are widely considered an effective approach to represent complex relationships [18]. A knowledge graph describes entities as **nodes** and their relations as **labeled edges**, serving as its two fundamental structures. Additionally, knowledge graphs enable the integration of more diverse data source into a single graph [16]. Ultimately, this paper hypothesizes that knowledge

graphs may support the development of cognitive maps due to their multiple structural similarities, as evidenced in the literature [8, 16]. Both represent entities and their relationships, enable navigation and inference, while organizing rich, interconnected information [1, 29]. Thus, we argue that knowledge graphs can enhance data integration and create a better understanding of complex structures through relational data modeling.

Development: In this paper, developing a web-optimized interface based on a knowledge graph approach serves as the design artifact solution. To build the artifact, *JavaScript* was selected as the core language due to its native performance optimization within modern web browsers and its rich ecosystem of high-performance rendering engines [9]. For the frontend architecture, *React* was chosen primarily for its robust state management capabilities and virtual *DOM* rendering [7]. Furthermore, to build the core graph-rendering engine, *Cytoscape.js* and *ReactFlow* were considered and experimented with. *Cytoscape.js* was used for the first prototype to quickly test graph layouts and basic user interactions with small datasets. Based on the comparisons addressed in Stage 4 (Section 3.4), we decided to use *ReactFlow* as the main library, due to its flexibility in styling and ease to debug. Finally, automotive configuration files in different formats (*ARXML*, *YAML*, and binary data) were integrated as datasets. A custom *Python* pipeline code was developed to parse, normalize, and convert these datasets into a unified *JSON* structure for the frontend knowledge graph, ultimately shown in Figure 2.

3.4 DSRM Stage 4 - Demonstration and Iteration

1st Iteration: The basic prototype was built with *Cytoscape.js* and *JavaScript* to visualize a small set of data with nodes and edges. However, due to the complexity and deeply nested structure of the data, the resulting visualization resembles a “snowball,” as illustrated in Figure 1-left. This snowball layout becomes increasingly difficult to manage when more data is added, as it is not possible to control the size of graph, demonstrated in Figure 1-center. Additionally, its circular, non-linear structure makes it hard for engineers to identify sequential data flow (starting and ending points), reducing overall comprehension, as illustrated in Figure 1-left.

2nd Iteration: The “snowball” visualization is transformed into a parallel-coordinates-style graph using *Cytoscape.js*. The main purpose of this iteration is to make the visualization easier to understand relationships, data flow sequences, and hierarchical structure between nodes, as shown in Figure 1-right. The prototype was shown to domain engineers for the first workshop feedback session. They confirmed the data relationships were correct but found the interface to not be intuitive. At the same time, due to graphical limitations and a lack of custom styling in *Cytoscape.js*, they claimed that the visualization was overly simple, failing to effectively communicate the complex structures. Also, they also mentioned that the difficulty of debugging with *Cytoscape.js* may cause maintenance issues in the future.

3rd Iteration: To overcome the styling and debugging problems identified in the first iteration, the rendering library was migrated to *ReactFlow*. *ReactFlow* was selected because its underlying data format is highly compatible with *Cytoscape.js*, requiring minimal structural translation. A key advantage of *ReactFlow* is that it allows

nodes and edges to be fully customized as interactive UI components, unlike *Cytoscape.js*, which renders them as SVG image elements. This helps build a better user-centered and intuitive design approach, as shown in Figure 2. For example, *ReactFlow* enables embedding rich UI elements such as buttons, collapsible panels, and formatted metadata directly inside each node, which is not easily achievable in *Cytoscape.js*. Finally, *ReactFlow* also improves debugging, since each node can be inspected directly in the browser developer tools.

4th Iteration: As the amount of data input increased on every iteration, the frontend started to have performance issues during rendering. Since the interface must compute hierarchical relationships and edge positions dynamically, the workload grew quickly and the interaction got overloaded. To address the challenges which were observed from another workshop feedback session with domain users, expensive calculations were wrapped in *React*’s *useMemo* hook, so they are only recomputed when the data changes. In addition, redundant parsing logic and duplicated code were refactored to improve performance and keep the code clean.

5th Iteration: Using the interface which had been developed through 4 rounds of iterations and based on user feedback over 2 workshop sessions, new features were added to enhance usability, including a search bar with auto-suggestion. This reduced typing effort and helped engineers quickly find deeply nested software modules using partial keywords. In addition, the interface was provided to different user groups with varying roles to identify their required data. Then relevant metadata was implemented accordingly to better support their tasks and responsibilities (Figure 2).

3.5 DSRM Stage 5 - Data Collection and Evaluation

Participant Profile: The proposed evaluation will involve domain experts from different engineering departments within the organization. This diverse selection of participants will ensure that the system is evaluated across a diverse range of automotive engineering workflows.

Experimental Tasks and Conditions Participants will be asked to complete a total of 9 distinct technical tasks across two environments: the current corporate interface (serving as the baseline control environment) and the newly developed knowledge-graph-based visualization interface. The 9 operational tasks require engineers to navigate real complex configuration data to find specific data values or components. Following the dual-process theory of cognitive science [23], 2 out of the 9 tasks are specifically designed to evaluate whether the relational layout of the new interface enhances the user’s inference abilities [8, 16], requiring active, contemplative, and high-effort mental processes (Type 2 processing [23]). In other words, these tasks are used to check whether the new interface helps users better understand complex data structures.

Quantitative Metrics: To evaluate performance efficiency, two objective quantitative metrics will be recorded during the testing sessions: **1) Completion Time:** The total duration (in seconds) required to successfully resolve each task; and **2) Interaction Frequency:** The total number of mouse clicks executed per task, serving as a measure for interaction and navigation overhead.

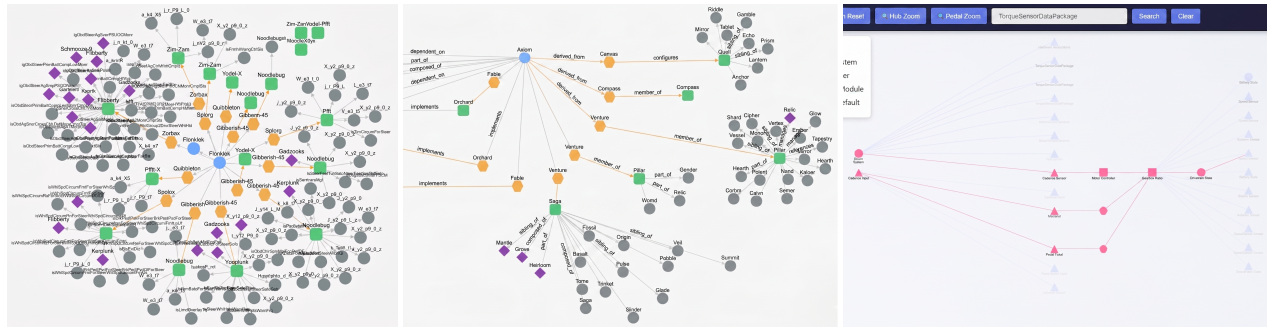


Figure 1: Left: The visualization resembles a snowball structure, which makes it difficult to extend as more data is added, resulting in uncontrolled growth in size. Center: The graph becomes increasingly difficult to interpret in terms of sequential and linear data flow, because of its non-linear structure. Right: Parallel coordinates-style graph shows sequential and hierarchical data structures.

Usability Test and Qualitative Survey: Participants will complete a post-test survey designed to capture both standardized usability metrics and subjective feedback. The survey instruments include: 1) **System Usability Scale (SUS):** A validated 10-item Likert-scale questionnaire used to calculate an overall usability score for both interfaces [4]; and 2) **Qualitative Feedback:** Open-ended textual feedback fields will be integrated to gather qualitative insights, user pain points, and feature requests regarding the layout and responsiveness of the artifact.

3.6 DSRM Stage 6 - Expected Outcomes

From the academic perspective, Human-centered computing provides a foundational understanding of the socio-technical relationship between data and humans. Design Science and DSRM present the importance of using the iterative development cycle to build design solutions. Additionally, as knowledge graphs and cognitive maps share strong similarities, we believe that building relationships between entities might support the formation of cognitive maps. However, to support this idea, direct measures of mental

workload, such as the NASA-TLX [13], or physiological methods [25] would need to be applied and evaluated in the experiment.

Thus, findings from interdisciplinary fields help bridge academia and industry. By integrating iterative user feedback into the development process, this approach ensures that the final artifact not only provides practical support for engineers working with complex vehicle data but also contributes broader theoretical insights into socio-technical system design. Finally, from an industrial perspective, iterative development helps guide future development of this sort of visualization. This includes scaling the system from a local file-based JSON setup to an enterprise database for real-time data handling. Besides the planned finalizing data collection and evaluation (Section 3.5), future work will also expand the interface to support different user groups with their different needs.

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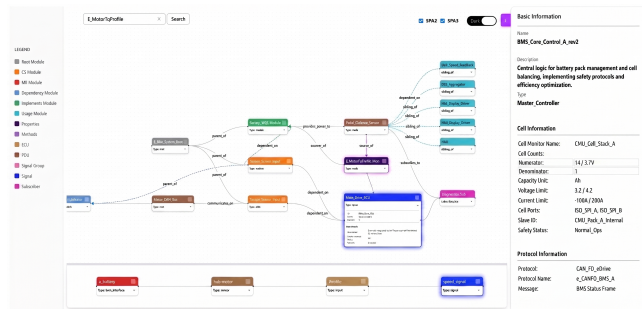


Figure 2: ReactFlow provides more intuitive design for graph visualization and user-centered interaction. User-specific data requirements are identified and translated into relevant metadata to support different operational tasks. AI Disclaimer: A generative AI model was used to digitally alter the names of the components in this image, in order to protect company property over the dataset.

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